A DECENT SPACE HAS FINITELY MANY IRREDUCIBLE COMPONENTS ABOVE ANY CLOSED POINT OF A NOETHERIAN BASE

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Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- Generic points exist. lemma-decent-generic-points (carded L26747). Generic points of the components.
- Finite fibre over closed points. lemma-universally-bounded-finite-fibres (same chapter, carded L26058). Finiteness of the fibre of the closed point.

Source reference. Original definition 03I8: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 1262–1262. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 03JV: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 732–732. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0BB8: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 4867–4867. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BAQ: focused-seeds/geometry/S9-decent-finitely-many-irreducible-components/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original theorem 086U: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 2323–2323. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. The following are equivalent*

Date: stacks-project:decent-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(1) *X is decent and $|X|$ has finitely many irreducible components,*
(2) *every quasi-compact scheme étale over X has finitely many irreducible components, there are finitely many $x \in |X|$ of codimension 0 on X, and each of these can be represented by a monomorphism $\mathrm{Spec}(k) \to X$,*
(3) *there exists a dense open $X' \subset X$ which is a scheme, X' has finitely many irreducible components with generic points $\{x'_1, \ldots, x'_m\}$, and the morphism $x'_j \to X$ is quasi-compact for $j = 1, \ldots, m$.*

Moreover, if these conditions hold, then X is reasonable and the points $x'_j \in |X|$ are the generic points of the irreducible components of $|X|$.

Proof. In the proof we use Properties of Spaces, Lemma 0BAQ without further mention. Assume (1). Then X has a dense open subscheme X' by Theorem 086U. Since the closure of an irreducible component of $|X'|$ is an irreducible component of $|X|$, we see that $|X'|$ has finitely many irreducible components. Thus (3) holds.

Assume $X' \subset X$ is as in (3). Let $\{x'_1, \ldots, x'_m\}$ be the generic points of the irreducible components of X'. Let $a : U \to X$ be an étale morphism with U a quasi-compact scheme. To prove (2) it suffices to show that U has finitely many irreducible components whose generic points lie over $\{x'_1, \ldots, x'_m\}$. It suffices to prove this for the members of a finite affine open cover of U, hence we may and do assume U is affine. Note that $U' = a^{-1}(X') \subset U$ is a dense open. Since $U' \to X'$ is an étale morphism of schemes, we see the generic points of irreducible components of U' are the points lying over $\{x'_1, \ldots, x'_m\}$. Since $x'_j \to X$ is quasi-compact there are finitely many points of U lying over x'_j (Lemma 03JV). Hence U' has finitely many irreducible components, which implies that the closures of these irreducible components are the irreducible components of U. Thus (2) holds.

Assume (2). This implies (1) and the final statement by Lemma 0BB8. (We also use that a reasonable algebraic space is decent, see discussion following Definition 03I8.) □